Full Stack Assignment – 06

Assignment ID: FSD-06

Title: Inventory Request Management System

Duration: 3 Hours

1️⃣ Objective

Design and develop an **Inventory Request Management System** that allows staff members to request items and administrators to manage inventory fulfillment.
The system must enforce quantity constraints and ensure **data persistence using a database**.

2️⃣ User Roles

🔷 STAFF

- Register and log in to the system
- View available inventory items
- Submit inventory requests
- View request status

🔷 ADMIN

- Log in to the system
- Manage inventory items
- View all inventory requests
- Approve or reject requests

3️⃣ Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** at API and UI levels

Inventory Management

- ADMIN can create and manage inventory items
- Inventory data must be stored in the database
- Available quantity must be fetched dynamically

Inventory Requests

- STAFF can request inventory items with quantity
- Requests must be stored in the database
- ADMIN can approve or reject requests

4 Business Rule (Critical)

⚠️ **Requested quantity must not exceed available inventory stock.**

Requests violating this rule must be rejected at the backend.

5 Technical Constraints (Mandatory)

- All inventory and request data **must persist in a database**
- Hardcoded or in-memory storage is **not allowed**
- Backend APIs must reflect real-time database state
- JWT authentication and role validation are mandatory
- Sensitive configuration must use environment variables

6 Frontend Requirements

- Login and registration pages
- Role-based dashboards for STAFF and ADMIN
- Inventory listing with available quantity
- Inventory request form
- Approve/Reject request actions for ADMIN

UI styling is not evaluated; correctness and clarity matter.

7 Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview
- Tech stack used
- User roles and permissions
- API endpoints
- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links

8️⃣ Evaluation Note

Evaluation focuses on **data validation, role enforcement, and end-to-end integration**.